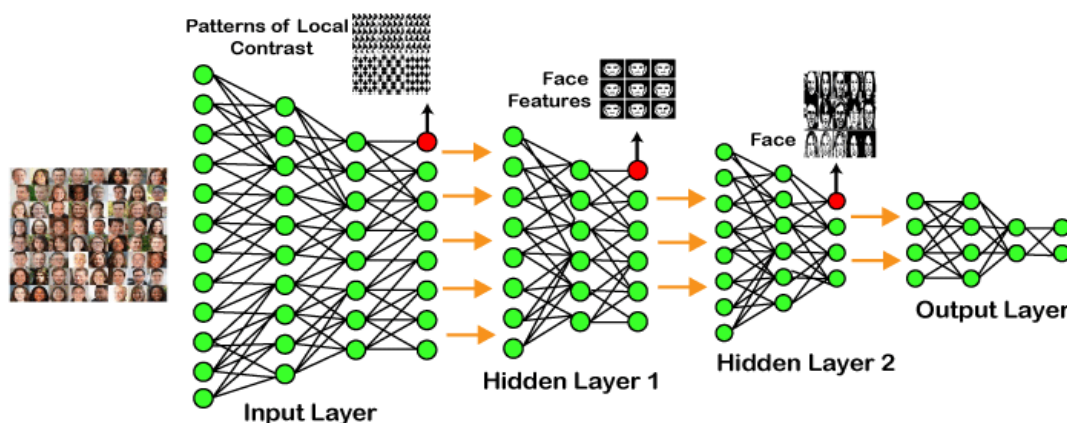


UNIT 6: DEEP LEARNING

Introduction to Deep Learning

- Deep learning is based on the branch of machine learning, which is a subset of artificial intelligence.
- Deep learning is a subset of machine learning in artificial intelligence (AI) that imitate the workings of the human brain in processing data and creating patterns for use in decision making.
- It involves neural networks with many layers ("deep") that can learn and make intelligent decisions on their own.
- Basically, it is a machine learning class that makes use of numerous nonlinear processing units so as to perform feature extraction as well as transformation.
- The output from each preceding layer is taken as input by each one of the successive layers.
- **[" Deep learning is a collection of statistical techniques of machine learning for learning feature hierarchies that are actually based on artificial neural networks. "]**

Example of Deep Learning



In the example given above, we provide the raw data of images to the first layer of the input layer. After then, these input layer will determine the patterns of local contrast that means it will differentiate on the basis of colors, luminosity, etc. Then the 1st hidden layer will determine the face feature, i.e., it will fixate on eyes, nose, and lips, etc. And then, it will fixate those face features on the correct face template. So, in the 2nd hidden layer, it will actually determine the correct face here as it can be seen in the above image, after which it will be sent to the output layer. Likewise, more hidden layers can be added to solve more complex problems, for example, if you want to find out a particular kind of face having large or light complexions. So, as and when the hidden layers increase, we are able to solve complex problems.

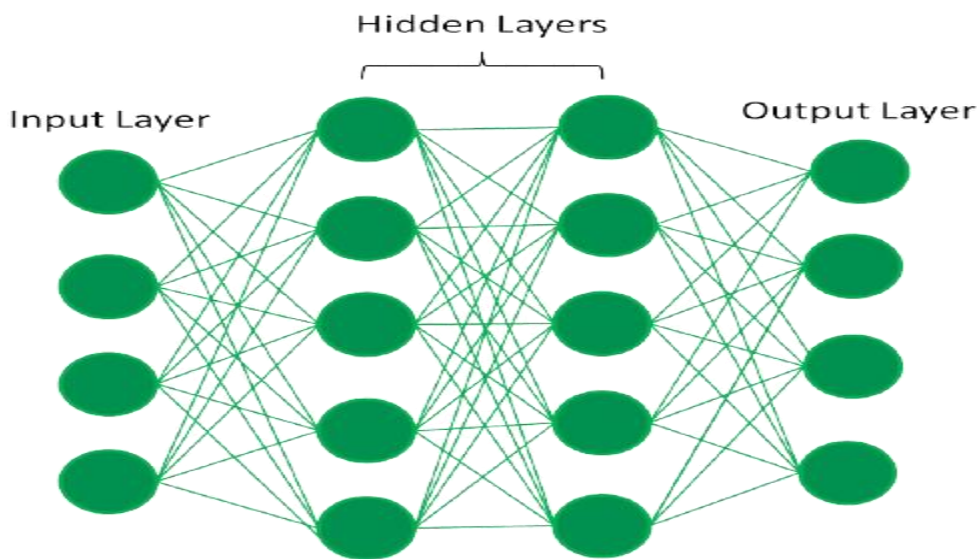
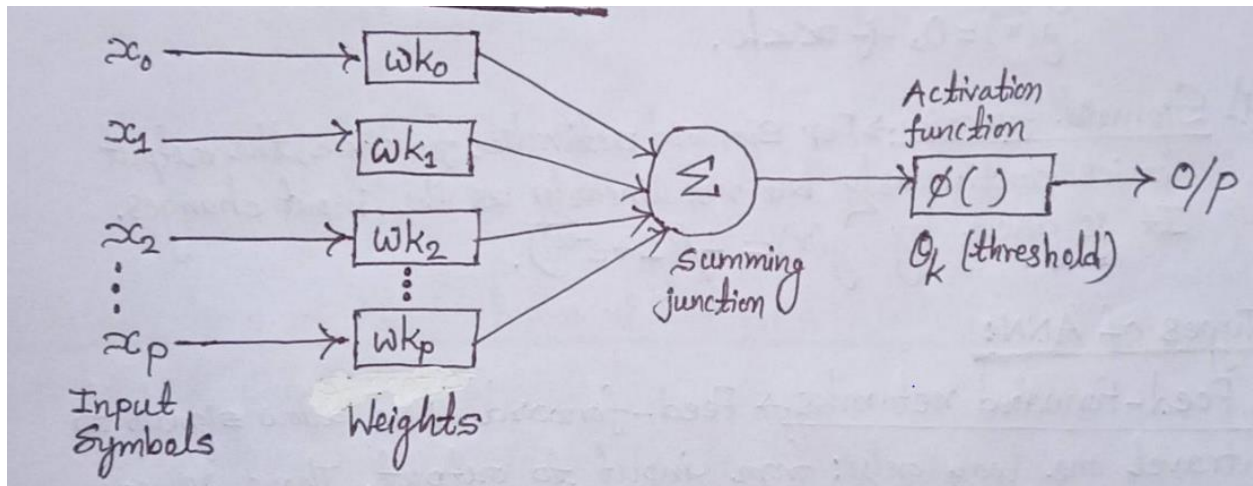
Deep learning applications

- **Self-Driving Cars:** In self-driven cars, it is able to capture the images around it by processing a huge amount of data, and then it will decide which actions should be incorporated to take a left or right or should it stop. So, accordingly, it will decide what actions it should take, which will further reduce the accidents that happen every year.
- **Voice Controlled Assistance:** When we talk about voice control assistance, then **Siri** is the one thing that comes into our mind. So, you can tell Siri whatever you want it to do it for you, and it will search it for you and display it for you.
- **Automatic Image Caption Generation:** Whatever image that you upload, the algorithm will work in such a way that it will generate caption accordingly. If you say blue colored eye, it will display a blue-colored eye with a caption at the bottom of the image.
- **Automatic Machine Translation:** With the help of automatic machine translation, we are able to convert one language into another with the help of deep learning.

Artificial Neural Network

- Artificial Neural Network is the non-linear, parallelly distributed and highly connected network having capacity of adaptability, self-organization, fault tolerance and very large scale integration (VLSI) implementation of neurons.
- Artificial neural networks are built on the principles of the structure and operation of human neurons.
- It is also known as neural networks or neural nets.
- An artificial neural network's input layer, which is the first layer, receives input from external sources and passes it on to the hidden layer, which is the second layer.
- Each neuron in the hidden layer gets information from the neurons in the previous layer, computes the weighted total, and then transfers it to the neurons in the next layer.
- These connections are weighted, which means that the impacts of the inputs from the preceding layer are more or less optimized by giving each input a distinct weight.
- These weights are then adjusted during the training process to enhance the performance of the model.
- The goal of ANN is to solve the problems in the same way that human mind does. ANN are widely used in computer vision, speech recognition, medical diagnoses, face recognition, signature verification etc.

Mathematical Model of ANN



Key components of ANNs

1. **Neurons:** The basic units of a neural network, similar to the biological neurons in the brain. Each neuron receives input, processes it and passes on the output to the next layer.
2. **Layers:**
 - **Input Layer:** The layer that receives the initial data.
 - **Hidden Layers:** Intermediate layers where the actual processing and feature extraction happen. These layers can vary in number and size.

- **Output Layer:** The final layer that produces the network's output.
- 3. **Weights and Biases:** Parameters within the network that are adjusted during training to minimize the error of the output.
 - **Weights:** Determine the importance of each input.
 - **Biases:** Allow the activation function to be shifted to the left or right, improving the model's fit.
- 4. **Activation Functions:** Functions that decide whether a neuron should be activated. Common activation functions include:
 - **Sigmoid:** Outputs a value between 0 and 1.
 - **Tanh:** Outputs a value between -1 and 1.
 - **ReLU (Rectified Linear Unit):** Outputs zero if the input is negative and outputs the input if it is positive.
 - **Softmax:** Used in the output layer for classification problems, converts raw scores to probabilities.

Types of neural networks

Deep Learning models are able to automatically learn features from the data, which makes them well-suited for tasks such as image recognition, speech recognition, and natural language processing. The most widely used architectures in deep learning are feedforward neural networks, convolutional neural networks (CNNs), and recurrent neural networks (RNNs).

1. **Feedforward neural networks (FNNs)** are the simplest type of ANN, with a linear flow of information through the network. FNNs have been widely used for tasks such as image classification, speech recognition, and natural language processing.
2. **Convolutional Neural Networks (CNNs)** are specifically for image and video recognition tasks. CNNs are able to automatically learn features from the

images, which makes them well-suited for tasks such as image classification, object detection, and image segmentation.

3. **Recurrent Neural Networks (RNNs)** are a type of neural network that is able to process sequential data, such as time series and natural language. RNNs are able to maintain an internal state that captures information about the previous inputs, which makes them well-suited for tasks such as speech recognition, natural language processing, and language translation.

Applications of deep learning

- **Image and speech recognition:** Identifying objects in images or converting spoken language into text.
- **Natural language processing (NLP):** Applications such as language translation, text summarization, and sentiment analysis.
- **Game playing:** AI systems like AlphaGo that can play and master games.
- **Autonomous systems:** Used in self-driving cars for tasks such as object detection and decision making.
- **Healthcare:** Predictive modeling for disease diagnosis, personalized treatment plans and medical image analysis.

Challenges in deep learning

Deep learning has made significant advancements in various fields, but there are still some challenges that need to be addressed. Here are some of the main challenges in deep learning:

1. **Data availability:** It requires large amounts of data to learn from. For using deep learning it's a big concern to gather as much data for training.

2. **Computational resources:** For training the deep learning model, it is computationally expensive because it requires specialized hardware like GPUs and TPUs.
3. **Time-consuming:** While working on sequential data depending on the computational resource it can take very large even in days or months.
4. **Interpretability:** Deep learning models are complex, it works like a black box. it is very difficult to interpret the result.
5. **Overfitting:** when the model is trained again and again, it becomes too specialized for the training data, leading to overfitting and poor performance on new data.

Advantages of deep learning:

1. **High accuracy:** Deep learning algorithms can achieve state-of-the-art performance in various tasks such as image recognition and natural language processing.
2. **Automated feature engineering:** Deep learning algorithms can automatically discover and learn relevant features from data without the need for manual feature engineering.
3. **Scalability:** Deep learning models can scale to handle large and complex datasets, and can learn from massive amounts of data.
4. **Flexibility:** Deep Learning models can be applied to a wide range of tasks and can handle various types of data, such as images, text and speech.
5. **Continual improvement:** Deep Learning models can continually improve their performance as more data becomes available.

Disadvantages of deep learning:

1. **High computational requirements:** Deep Learning AI models require large amounts of data and computational resources to train and optimize.

2. **Requires large amounts of labeled data:** Deep Learning models often require a large amount of labeled data for training, which can be expensive and time-consuming to acquire.
3. **Interpretability:** Deep Learning models can be challenging to interpret, making it difficult to understand how they make decisions.
Overfitting: Deep learning models can sometimes overfit to the training data, resulting in poor performance on new and unseen data.
4. **Black-box nature:** Deep learning models are often treated as black boxes, making it difficult to understand how they work and how they arrived at their predictions.

Bayesian learning and its Applications

- Bayesian learning is an approach to machine learning based on Bayes' theorem, which provides a principled way of updating beliefs in the light of new evidence.
- It treats model parameters as random variables and incorporates prior knowledge along with observed data to make predictions or infer parameters.
- Bayesian learning provides a robust framework for making inferences and predictions particularly when dealing with uncertainty and incorporating prior knowledge.
- It continues to be an important area of research and application in machine learning and statistics.

Bayes Theorem

Bayes theorem is a fundamental concept in probability theory that plays a crucial role in various machine learning algorithms, especially in the fields of Bayesian statistics and probabilistic modelling. It provides a way to update probabilities based on new evidence or information. In the context of machine learning, Bayes' theorem is often used in Bayesian inference and probabilistic models.

The theorem can be mathematically expressed as:

$$P(A|B) = P(B|A) \cdot P(A) / P(B)$$

where,

- $P(A|B)$ is the posterior probability of event A given event B.
- $P(B|A)$ is the likelihood of event B given event A.
- $P(A)$ is the prior probability of event A.
- $P(B)$ is the total probability of event B.

Key concepts

1. **Prior distribution:** Represents the initial belief about the parameters before observing any data.
2. **Likelihood:** The probability of the observed data given the parameters.
3. **Posterior distribution:** The updated belief about the parameters after considering the observed data.
4. **Marginal likelihood (Evidence):** The probability of the data under all possible parameter values.

Bayesian inference

Bayesian inference is a method of statistical inference in which Bayes' Theorem is used to update the probability of a hypothesis as more evidence or information becomes available.

The goal of Bayesian inference is to update the posterior distribution $P(A|B)$ using Bayes theorem. This involves:

1. **Choosing a Prior:** Selecting an appropriate prior distribution $P(A)$ based on prior knowledge or assumptions.
2. **Calculating the Likelihood:** Determining the likelihood function $P(B|A)$ based on the model and observed data.

3. **Computing the Posterior:** Using Bayes' theorem to combine the prior and likelihood to get the posterior distribution.

Applications

1. **Regression and Classification:** Bayesian methods can be applied to both regression and classification problems, providing probabilistic predictions and uncertainty estimates.
2. **Time Series Analysis:** Bayesian approaches are used for forecasting and analyzing temporal data, capturing uncertainty in predictions.
3. **Natural language processing:** Techniques like Latent Dirichlet Allocation (LDA) for topic modeling are based on Bayesian principles.
4. **Recommender systems:** Bayesian methods are used to model user preferences and provide personalized recommendations.
5. **Medical diagnosis:** Bayesian inference is employed to combine prior knowledge with patient data for diagnosis and treatment planning.

Advantages and Challenges

Advantages:

- Incorporates prior knowledge and provides a principled way to update beliefs with new data.
- Offers probabilistic interpretations, allowing for uncertainty quantification.
- Flexibility in modeling complex problems and incorporating different types of data.

Challenges:

- Computationally intensive, especially for large datasets and complex models.
- Sensitivity to the choice of prior, which can influence results.
- Requires expertise in probability and statistical modeling.